Relative Performance Markets: Extending Prediction Markets and Derivatives Through Comparative Outcome Trading

Abstract

Financial markets provide mature mechanisms for trading assets, transferring risk, and expressing beliefs about future events. Prediction markets allow participants to trade contingent claims whose values depend on uncertain outcomes, while derivatives allow exposure to be created around the future value or behavior of an underlying variable. Both structures have proven capable of turning expectations into prices. Yet most such markets are constructed around either an absolute outcome or the value of an underlying asset.

This paper proposes **Relative Performance Markets (RPMs)** as a distinct market design framework in which the economically relevant outcome is not whether an entity succeeds in absolute terms, but how its measured performance compares with a defined reference set or benchmark.

Instead of asking whether A will increase, an RPM asks whether A will perform better or worse than its reference environment. A participant can therefore express a view on A without requiring the opposing participant to explicitly take the inverse view on A. Exposure may instead emerge through allocations across competing entities within a common system.

The concept draws from prediction markets, contingent claims, derivatives, pari-mutuel systems, relative-performance evaluation, and benchmark-based finance while differing from each. The paper develops the conceptual foundations of RPMs, introduces a general mathematical representation, examines closed-pool and benchmark-relative formulations, distinguishes measured performance from market expectations, and identifies the principal problems that must be solved before such markets can be considered financially complete.

The central proposition is that **relative performance itself can serve as a tradable underlying**.

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1. Introduction

Markets have developed many ways to trade uncertainty.

An investor may buy an equity because they expect a company to become more valuable. A derivatives trader may obtain exposure to interest rates, inflation, commodities, volatility, or credit events without owning the underlying economic object. A prediction-market participant may purchase a contingent claim because they believe a particular event is more likely than the market price implies.

These mechanisms differ substantially, but they share a common idea: uncertain future states can be transformed into financial claims.

Prediction markets are particularly interesting because they demonstrate that markets can function not only as mechanisms for transferring risk but also as mechanisms for aggregating dispersed information. Wolfers and Zitzewitz describe prediction markets as markets capable of aggregating dispersed information into forecasts, with appropriately designed contracts revealing expectations about probabilities and other properties of uncertain outcomes. (aeaweb.org) Experimental work on Arrow-Debreu-style markets similarly provides evidence that experienced participants can cause prices to converge toward information contained across traders, although such aggregation is not automatic and can perform poorly in difficult or unfamiliar environments. (link.springer.com)

Derivatives extend a different principle: ownership of an underlying object is not necessary to create economically meaningful exposure to it. Contracts can instead define payments according to observable future states.

This principle has already been extended to macroeconomic information. In 2002, Deutsche Bank and Goldman Sachs introduced auctions for derivatives based directly on macroeconomic releases such as U.S. nonfarm payrolls. Early auctions created millions of dollars of notional exposure around the value of an official statistical release. (goldmansachs.com) Subsequent academic analysis found that market-derived expectations from these economic derivatives were comparable with survey forecasts and in some respects better reflected subsequent financial-market reactions to macroeconomic surprises. (nber.org)

These developments establish two important precedents:

1. information about uncertain future states can be priced by markets; and
2. externally measured data can serve as the settlement reference for financial claims without representing a conventionally ownable asset.

This paper considers a third possibility.

What if the underlying object is not an absolute event or absolute measurement, but **relative performance**?

Consider a set of measurable entities:

$$\mathcal{E} = \{E_1, E_2, \dots, E_n\}.$$

Each entity produces some measurable performance over time. Instead of creating independent markets asking whether each entity will rise or fall, the system constructs a reference from the population and expresses each entity relative to that reference.

The fundamental question becomes:

How is E_i performing relative to its relevant world?

This creates what this paper calls a **Relative Performance Market**.

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2. From Absolute Outcomes to Relative Outcomes

Suppose two companies each increase revenue during a year.

Company A grows by 8%.

Company B grows by 20%.

An absolute proposition about Company A may classify the year positively because A grew.

A relative proposition can simultaneously state that A underperformed its reference environment.

Neither statement is contradictory.

This distinction appears throughout economics and finance. Investment performance is routinely evaluated against benchmarks. Relative-performance evaluation has also been studied in compensation and contracting because comparison with peers can remove common shocks that affect all participants. Gibbons and Murphy note that relative-performance evaluation can partially insulate measured performance from common uncertainty affecting multiple agents, although it introduces its own strategic problems. (nber.org) Benchmark selection itself is consequential: empirical work on fund evaluation has shown that measured relative performance can vary materially depending on the benchmark selected. (nber.org)

Relative measurement therefore already exists.

What is less common is treating the resulting **relative state itself as the underlying of a continuously tradable market**.

An RPM formalizes this distinction.

Let the measured performance state of entity i at time t be:

$$S_i(t).$$

Let the relevant benchmark be:

$$B(t).$$

A simple relative state may be represented as:

$$R_i(t) = S_i(t) - B(t).$$

Alternatively, where the underlying state supports ratios:

$$R_i(t) = \frac{S_i(t)}{B(t)}.$$

The precise transformation is application-dependent.

The essential property is not subtraction or division. It is that the financial proposition concerns:

$$E_i \text{ relative to } B$$

rather than:

$$E_i \text{ in isolation.}$$

This apparently small transformation has substantial consequences for market structure.

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3. Prediction Markets as a Starting Point

Prediction markets provide the most intuitive starting point.

Consider a conventional binary market:

> Will event X occur before date T?

Two complementary claims may exist:

$$YES + NO = 1.$$

At settlement:

$$YES = \begin{cases} 1 & \text{if } X \text{ occurs} \\ 0 & \text{otherwise} \end{cases}$$

and:

$$NO = 1 - YES.$$

Before settlement, the claims trade.

Under simplified assumptions, the market price of YES may be interpreted as a market-implied probability. More generally, prediction-market prices represent market valuations of state-contingent claims and may aggregate information distributed among participants. Research shows that this information aggregation can be powerful, but depends on security design, participant behavior, information structure, and market experience. (aeaweb.org)

The important structural property is **complementarity**.

If a complete set of claims is created for $1, the system does not need to manufacture additional dollars when the outcome becomes known.

One claim wins because another loses.

This produces a conservation relationship:

$$\sum_{k=1}^{m} P_k(T) = 1$$

for mutually exclusive and collectively exhaustive terminal states.

Relative Performance Markets borrow this idea but relax the requirement that the state space be a simple binary event.

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4. Beyond Binary Prediction

Many economically meaningful questions are continuous.

Inflation is not naturally YES or NO.

GDP growth is not naturally YES or NO.

Temperature is not naturally YES or NO.

Company growth is not naturally YES or NO.

One can artificially convert these into binary propositions:

> Will inflation exceed 3%?

but doing so discards information.

A continuous market instead asks:

> Where will the measured value finish?

Macroeconomic derivatives demonstrated that financial claims could be constructed directly around numerical economic releases. The economic-derivatives markets launched in 2002 included contracts around nonfarm payrolls and subsequently other scheduled macroeconomic indicators. (goldmansachs.com)

Gürkaynak and Wolfers analyzed such markets and showed that their prices could produce distributions over future macroeconomic outcomes rather than merely point forecasts. (nber.org)

Relative Performance Markets take another step.

Rather than settling only against:

$$X_T$$

they can settle against:

$$X_T - B_T.$$

The underlying uncertainty becomes **comparative**.

A market may therefore ask:

> Where will A finish relative to its benchmark?

rather than:

> Where will A finish?

This is not merely a semantic difference.

The benchmark can remove common movement and transform the economic meaning of the claim.

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5. Common Shocks

Suppose:

$$A = +5\%$$

and:

$$B = +8\%.$$

A performed positively in absolute terms but negatively in relative terms:

$$R_A = 5\% - 8\% = -3\%.$$

Now suppose instead:

$$A = -2\%$$

and:

$$B = -10\%.$$

A declined in absolute terms but substantially outperformed its environment:

$$R_A = -2\% - (-10\%) = +8\%.$$

An RPM therefore distinguishes:

performance

from

relative performance.

This can be useful when common shocks affect an entire population.

The idea has parallels in tournament theory and relative-performance contracting. Green and Stokey showed that rank- or relative-based compensation can become valuable where common uncertainty affects participants, because comparison can help remove a common shock. (nber.org)

An RPM applies the comparative principle to market exposure rather than employee compensation.

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6. A General Relative Performance Framework

Consider N measurable entities.

Each entity has a normalized state:

$$S_i(t), \qquad i \in \{1, \dots, N\}.$$

Define a benchmark function:

$$B(t) = F(S_1(t), S_2(t), \dots, S_N(t)).$$

The simplest possible benchmark is an arithmetic mean:

$$B(t) = \frac{1}{N} \sum_{i=1}^{N} S_i(t).$$

A relative state becomes:

$$R_i(t) = S_i(t) - B(t).$$

More generally:

$$R_i(t) = G(S_i(t), B(t))$$

for some transformation G.

This definition immediately introduces one of the central design decisions of an RPM:

What constitutes the appropriate benchmark?

The answer cannot be treated as incidental.

If entities are weighted according to size,

$$B(t) = \sum_i w_i S_i(t), \qquad \sum_i w_i = 1,$$

then large entities exert greater influence on the benchmark.

If entities are equally weighted,

$$w_i = \frac{1}{N},$$

then each entity contributes equally.

If the benchmark excludes the entity being measured:

$$B_{-i}(t) = \frac{1}{N-1} \sum_{j \neq i} S_j(t),$$

then the relative state:

$$R_i(t) = S_i(t) - B_{-i}(t)$$

eliminates self-inclusion.

None of these constructions is universally correct.

The benchmark defines the meaning of relative performance.

This is consistent with evidence from conventional benchmark evaluation: conclusions about relative performance can change when the benchmark changes. (nber.org)

Therefore benchmark construction is part of the market's economic specification, not merely an implementation detail.

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7. Relative Performance Markets Versus Prediction Markets

RPMs share several properties with prediction markets.

Both allow participants to express beliefs about uncertain future states.

Both can transform dispersed expectations into prices.

Both may ultimately settle against an externally verifiable outcome.

Both can be designed around fully collateralized contingent claims.

However, the information being traded differs.

A conventional prediction market may ask:

$$P(X)$$

where X is an event.

A relative performance market concerns something closer to:

$$P(S_A - S_B)$$

or, more generally:

$$P(S_i - B).$$

Furthermore, the benchmark itself may be dynamic.

This creates an important distinction.

In an RPM, an entity's relative position may change even when no new information directly concerning that entity has arrived.

Suppose:

$$S_A(t+1) = S_A(t)$$

but:

$$B(t+1) > B(t).$$

Then:

$$R_A(t+1) < R_A(t).$$

A's relative position has deteriorated because its environment improved.

This means the state of an RPM can be inherently interconnected.

Independent prediction markets generally do not possess this property.

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8. Relative Performance Markets Versus Derivatives

Derivatives provide the second conceptual ancestor.

A derivative creates a payoff dependent on some underlying reference.

That reference does not have to be physically ownable.

Interest rates, volatility, weather conditions, credit events, and economic statistics can all support contingent contracts.

Economic derivatives are particularly relevant because they demonstrate direct exposure to official economic data rather than exposure through traditional proxies. The original 2002 economic-derivative auctions were explicitly presented as a mechanism for obtaining direct exposure to macroeconomic releases. (goldmansachs.com)

An RPM can similarly define a terminal payoff:

$$P_i(T) = H(R_i(T))$$

where $R_i(T)$ is the verified relative performance of entity i.

However, conventional derivatives frequently require bilateral counterparties, clearing arrangements, margin, or market makers willing to warehouse risk.

If aggregate participants demand a large amount of positive exposure to one outcome, someone must ultimately absorb the corresponding negative exposure.

This produces the underwriting problem:

Who pays the winner?

An RPM need not eliminate this question.

Instead, one possible RPM design changes how it is answered.

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9. Closed-Pool Relative Performance

Consider a fixed collateral pool:

$$C.$$

Participants allocate claims across N entities.

Let:

$$V_i(t)$$

represent the aggregate redeemable value associated with entity i.

A fully conserved system would maintain:

$$\sum_{i=1}^{N} V_i(t) = C$$

ignoring fees and external cash flows.

Relative performance would redistribute claim value rather than create new aggregate liabilities.

If entity A gains \$10 of redeemable value, the rest of the system must collectively lose \$10.

Thus:

$$\sum_i \Delta V_i = 0.$$

This is conceptually closer to a pari-mutuel system than to a conventional leveraged derivative book.

In pari-mutuel markets, participants' wagers form a pool and the payoff to winning positions depends partly on how capital has been distributed across

outcomes. Research on pari-mutuel markets emphasizes this endogenous relationship between bets, implied odds, and eventual payouts. (sciencedirect.com)

Economic derivatives themselves have historically experimented with related mechanisms. The early macroeconomic-derivative auctions used pari-mutuel techniques intended to match collected premiums with possible payouts. (nber.org)

This precedent is significant.

It demonstrates that the combination of:

- economic information,
- derivative-like contingent exposure, and
- mutualized risk

is not without historical antecedent.

The proposed RPM generalization is different, however, because relative performance can involve multiple continuously evolving entities rather than a single scheduled data release.

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10. The Conservation Problem

Closed collateral imposes a mathematical restriction that is easy to overlook.

Suppose:

$$\$80$$

is allocated to A and:

$$\$20$$

to B.

Assume an external performance measurement reports:

$$A = +4\%$$

and:

$$B = -1\%.$$

It is impossible for a closed $100 system simply to promise every A holder +4% and every B holder -1%.

A would require:

$$80(0.04) = \$3.20$$

of gains.

B would surrender only:

$$20(0.01) = \$0.20.$$

The system would therefore owe an additional:

$$\$3.00.$$

That money does not exist.

Consequently, in a conserved RPM:

$$\text{measured relative performance} \neq \text{automatic monetary return.}$$

This distinction is foundational.

A performance signal can determine the direction and magnitude of economic information without mechanically determining dollar P&L.

Monetary revaluation must also account for the distribution of capital.

A general mechanism therefore requires something resembling:

$$V_i(t+1) = C\frac{A_i(t)\phi(R_i(t+1))}{\sum_j A_j(t)\phi(R_j(t+1))}$$

where:

- $A_i(t)$ represents some measure of claims or allocation to entity i;
- R_i represents measured relative performance; and
- ϕ transforms performance into a positive weighting function.

This equation is illustrative rather than proposed as a final RPM mechanism.

Its purpose is to demonstrate the conservation requirement:

$$\sum_i V_i(t+1) = C.$$

Any production RPM mechanism must additionally solve path dependence, entry pricing, withdrawal, switching, rounding, settlement, and strategic behavior.

Those problems remain open in this paper.

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11. Crowding as an Economic Property

Conservation produces an interesting consequence.

Suppose nearly everyone correctly believes A will outperform.

If 99% of collateral is already allocated to A, there is little remaining capital from underperforming allocations available for redistribution toward A.

Thus being correct about performance does not necessarily imply a large return.

The market must also ask:

> **How much of this belief is already represented in positioning?**

This resembles the logic of market prices generally: identifying the winner is insufficient if everyone already expects the same winner.

In an RPM, however, this property can arise directly from conservation.

An under-owned entity that substantially outperforms may generate greater redistribution per unit of allocated capital than an extremely crowded entity producing the same measured performance.

Consequently:

$$\text{return} = f(\text{relative performance}, \text{entry state}, \text{capital distribution}).$$

This creates endogenous pricing pressure without requiring the performance index itself to encode market positioning.

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12. Economic State and Market Price Must Be Distinct

A Relative Performance Market requires at least two conceptually different numbers.

12.1 Reference state

The reference state describes what verified measurements currently imply:

$$R_i(t).$$

12.2 Market price

The market price describes what participants are willing to pay for exposure given expectations about the future:

$$M_i(t).$$

These need not be equal.

Suppose A's current verified relative state is:

$$R_A = 105.$$

The market may price its terminal state as though it expects:

$$110.$$

Later, new evidence moves the verified state to:

$$107.$$

A has objectively improved.

Yet if market expectations simultaneously fall from 110 to 108, the traded claim can decline in price.

Therefore:

An entity can improve while its RPM market price falls.

This is not a defect.

It is the distinction between **realized information** and **expected future information**.

Prediction markets exhibit an analogous distinction: the occurrence of favorable intermediate evidence need not raise a contract's price if the information was already anticipated or if other evidence changes the expected terminal outcome.

An RPM should therefore avoid defining its market price as merely the latest observed performance number.

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13. Continuous Markets and Finite Settlement

A useful RPM may trade continuously while settling over finite horizons.

Let an epoch cover:

$$[t_0, T].$$

At t_0, the performance reference is initialized.

During the epoch:

1. measurements arrive;
2. reference states evolve;
3. participants update expectations;
4. claims trade;
5. positions may be entered and exited.

At T, a predefined methodology determines the final verified relative state.

The epoch then settles.

This structure combines continuous price discovery with finite liability.

It also prevents a difficult problem associated with perpetual contingent claims: determining what ultimately anchors market value.

Finite settlement creates convergence toward an externally determined terminal reference.

The consumer-facing market may nevertheless appear persistent if successive epochs are automatically rolled.

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14. Entry After Performance Has Already Occurred

Continuous participation creates another essential requirement.

Suppose A begins an epoch at:

100

and reaches:

110.

A new participant entering at 110 must not receive the economic benefit of the move from 100 to 110.

Otherwise participants could wait until performance had already occurred and purchase historical gains.

Therefore positions require an entry state or market acquisition price.

If a participant enters at:

$$M_A(t_e),$$

their economic exposure begins at t_e, not at the beginning of the epoch.

This sounds trivial, but it becomes difficult in a closed-pool mechanism because deposits and withdrawals change the amount of capital participating in future redistribution.

A valid RPM must ensure that:

- existing holders retain previously earned value;
- new entrants do not capture past redistribution;
- exiting holders do not extract future value;
- remaining claims stay fully collateralized.

This is one of the central mathematical problems left for subsequent mechanism design.

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15. Relative Markets as Networks

A further property emerges when many entities share the same benchmark.

Suppose:

$$A/B = \frac{A/W}{B/W}$$

where W is a common benchmark.

If every entity has a canonical relative representation against W, pairwise relative views may potentially be derived without constructing an entirely independent market for every pair.

For example:

$$A/W = 1.08$$

and:

$$B/W = 1.03.$$

Then:

$$A/B = \frac{1.08}{1.03} \approx 1.0485.$$

This resembles cross-rate construction in foreign-exchange markets.

A common benchmark therefore acts as a coordinate system.

With N entities, the system may represent a large number of comparative relationships from N canonical benchmark relationships rather than requiring:

$$\frac{N(N-1)}{2}$$

independent markets.

Whether this mathematical relationship can also support efficient financial netting depends on the eventual claim and collateral mechanism and should not be assumed merely from the arithmetic identity.

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16. Relative Performance Versus Ranking

An RPM should not be confused with a tournament or leaderboard.

Ranking asks:

> Who is first?

Relative performance asks:

> By how much did A outperform or underperform its reference?

Suppose:

$$A = +10\%, \quad B = +9.9\%, \quad C = -20\%.$$

A ranking gives:

$$A > B > C.$$

But that discards the magnitude of the differences.

An RPM can preserve continuous relative distance.

This is important because a market needs more information than ordinal rank if its claims are to respond proportionally to measured performance.

Nevertheless, tournament theory remains informative because it demonstrates both the benefits and dangers of relative evaluation. Relative measurement can filter common shocks, but it can also alter incentives, encourage strategic behavior, and create attempts to influence peers rather than improve one's own performance. (nber.org)

These concerns become particularly important when the entities being measured can react to the benchmark.

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17. Information Aggregation

Prediction markets are often valued because prices can aggregate dispersed beliefs.

An RPM potentially performs a similar function but around comparative beliefs.

Participants may possess information about:

- entity A;
- competing entities;
- the benchmark;
- common shocks;
- correlations;
- future measurements;
- how other participants interpret those measurements.

A market price for A/B can therefore aggregate information not merely about A but about A's expected performance relative to B.

This makes relative markets potentially information-dense.

However, the literature warns against assuming perfect aggregation.

Experimental evidence indicates that market participants may initially fail to aggregate complicated information effectively, with performance improving as participants gain experience. (link.springer.com) Theoretical work similarly establishes conditions under which some securities aggregate information while others may fail to do so. Ostrovsky's work distinguishes "separable" securities for which information aggregation has strong theoretical properties from non-separable securities where aggregation may fail under some information structures. (doi.org)

Consequently, RPM design cannot simply assume:

$$\text{market price} = \text{true expectation}.$$

Contract structure matters.

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18. Market Scoring Rules and Automated Liquidity

Prediction markets have also developed mechanisms that allow markets to operate without requiring a traditional order book with a willing counterparty at every instant.

Hanson's logarithmic market scoring rule provides one influential example, combining scoring-rule elicitation with market-based information aggregation. (ubplj.org)

Such mechanisms are relevant to RPM research because relative markets face a fundamental liquidity question:

> If everyone wants exposure to the same entity, who takes the other side?

There are several broad possibilities:

1. traditional bilateral counterparties;
2. external market makers;
3. automated market makers;
4. subsidized market scoring rules;
5. complete-set issuance;
6. pari-mutuel or closed-pool redistribution;
7. hybrid structures.

Each has different consequences.

A subsidized market maker may provide liquidity but assumes bounded or unbounded loss depending on design.

A bilateral derivative requires counterparties.

A closed pool conserves collateral but makes payouts dependent on allocation.

Complete sets provide strong collateral accounting but can suffer from demand imbalance between complementary claims.

No mechanism should be assumed superior without formal analysis.

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19. Why Relative Performance May Reduce the Counterparty Problem

Consider independent markets:

A will rise.

B will rise.

C will rise.

It is possible for participants to demand long exposure to all three simultaneously.

Someone must provide the corresponding short exposure.

Now consider:

A relative to the population.

B relative to the population.

C relative to the population.

Relative performance creates natural opposition.

If A outperforms the aggregate, some combination of other entities must mathematically underperform the aggregate.

For a simple arithmetic benchmark:

$$B = \frac{1}{N} \sum_i S_i,$$

we have:

$$\sum_i (S_i - B) = 0.$$

Therefore:

$$\sum_i R_i = 0.$$

Relative performance is naturally zero-centered.

This is a powerful property.

It means the **measurement system itself contains opposition**.

However, this does not automatically prove that financial liabilities are zero-sum, because capital may be unevenly distributed.

The earlier example demonstrates why:

$$\sum_i R_i = 0$$

does not imply:

$$\sum_i A_i R_i = 0.$$

Therefore relative measurement helps structurally, but a financial mechanism is still required to translate zero-centered performance into conserved monetary claims.

This distinction is critical.

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20. A Closed-Loop Interpretation

The strongest conceptual form of an RPM is therefore a **closed-loop market**.

Participants contribute collateral to a shared system.

They allocate capital according to beliefs about relative future performance.

Measured outcomes determine which allocations outperform.

Market positioning determines how valuable it was to be correct.

Aggregate collateral is conserved.

Conceptually:

capital → relative allocations → measured performance → redistribution → new allocations.

No external sponsor must necessarily create the winner's gain.

Instead:

relative underperformance funds relative outperformance.

This is conceptually related to pari-mutuel betting, but unlike a conventional race, the subjects may be continuously measured, outcomes may be continuous rather than discrete, and participants may enter and exit before terminal settlement.

Those differences are sufficiently substantial that existing pari-mutuel theory cannot simply be imported unchanged.

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21. What Exactly Is Being Traded?

This question deserves a precise answer.

An RPM participant is not necessarily purchasing ownership in entity A.

They are not necessarily lending to A.

They may have no legal claim against A.

Instead, they acquire a financial claim whose value depends on:

measured performance of A relative to benchmark B.

Thus the underlying is informational.

This has precedent.

Macroeconomic derivatives demonstrated that published statistics can themselves serve as financial reference variables. (nber.org)

An RPM generalizes the idea:

> **measured comparative performance can serve as a financial reference variable.**

The asset being traded is therefore not A itself.

It is **exposure to A's relative measured outcome**.

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22. Objective Measurement and Oracle Dependence

A Relative Performance Market cannot function without a credible answer to:

> What was the performance?

For a sporting event, observation is comparatively simple.

For economic or social variables, measurement may be significantly more complicated.

Let raw observations be:

$$D(t) = \{d_1, d_2, \dots, d_m\}.$$

A measurement methodology:

$$\mathcal{M}$$

produces:

$$S_i(t) = \mathcal{M}(D_i(t)).$$

The market therefore inherits the assumptions embedded in $\mathcal{M}$.

If the methodology is discretionary, participants must trust whoever controls it.

If it is deterministic and public, participants can independently reproduce the reference state, but they must still trust or verify the underlying observations.

Therefore RPMs introduce two separate truth problems:

Evidence truth

Was the underlying observation correctly reported?

Methodological truth

Does the agreed methodology transform those observations exactly as specified?

These should be separated.

A robust system should minimize discretionary intervention after positions have been created.

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23. Measurement Is Not Price

This separation also clarifies the role of the market.

The measurement system should answer:

> What does verified evidence currently imply?

The market should answer:

> What do participants believe the terminal relative state will be?

These functions should not be collapsed.

If a centralized administrator can both determine the measurement and control market pricing, the informational value of the market is compromised.

Conversely, if market price itself determines settlement truth, the system becomes reflexive.

A clean architecture therefore separates:

$$\text{evidence} \rightarrow \text{reference state}$$

from:

$$\text{beliefs} \rightarrow \text{market price}.$$

Settlement reconnects them at a predefined terminal point.

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24. Applications Beyond Financial Assets

Relative Performance Markets are not inherently limited to national economies or conventional financial objects.

In principle, any domain satisfying several conditions could support an RPM:

1. multiple comparable entities exist;
2. performance can be measured;
3. a defensible benchmark can be constructed;
4. measurements can be independently verified;
5. participants have heterogeneous expectations about future relative performance.

Potential research domains include:

- companies relative to sectors;
- cities relative to peer cities;
- energy systems relative to regional benchmarks;
- logistics networks;
- agricultural regions;
- technology ecosystems;
- environmental performance;
- industrial sectors;
- public infrastructure systems.

This does not imply that all such markets would be legally, ethically, economically, or practically desirable.

It demonstrates that the primitive is more general than any single application.

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25. The Benchmark as a Numeraire

The benchmark may play a deeper role than merely providing a subtraction term.

If every entity is represented relative to the same reference W, then W becomes a type of performance numeraire.

Prices may be represented as:

$$E_1/W, \quad E_2/W, \quad \dots, \quad E_N/W.$$

Cross-performance can then be derived:

$$E_i/E_j = \frac{E_i/W}{E_j/W}.$$

This creates an analogy with currency markets, where a common numeraire permits cross-rates to be calculated.

The analogy should not be taken too far.

Currencies have monetary functions and institutional structures that a performance benchmark need not possess.

Nevertheless, mathematically, a shared benchmark provides a **common coordinate system for relative performance**.

This may substantially reduce fragmentation in large RPM systems.

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26. Benchmark Reflexivity

If entity A contributes to benchmark W and is simultaneously measured against W, A partly influences its own reference.

For:

$$W = \sum_i w_i S_i,$$

the relative value:

$$S_A - W$$

contains S_A on both sides.

This is particularly significant where:

$$w_A$$

is large.

Possible approaches include:

Small equal weights

With sufficiently many equally weighted entities, any individual contribution becomes small.

Ex-entity benchmark

Construct:

$$W_{-i}$$

for each entity.

Fixed external benchmark

Use a reference not constructed from participating entities.

Hierarchical benchmarks

Compare entities first against peers and subsequently aggregate peer groups.

Each approach changes the economic meaning of the market.

There is no universally correct solution.

Benchmark reflexivity should therefore be considered a first-class design problem rather than an implementation inconvenience.

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27. Dynamic Membership

What happens when an entity enters or leaves the benchmark?

Suppose:

$$W_t = \frac{1}{N} \sum_i S_i(t)$$

and a new entity is added.

The benchmark may change even though no underlying entity experienced new performance.

This creates artificial movement.

A mature RPM therefore requires membership rules addressing:

- admission;
- removal;

- missing data;
- suspended entities;
- benchmark reconstitution;
- historical continuity;
- normalization at entry;
- treatment of failed or unmeasurable entities.

Traditional financial indices already demonstrate that benchmark methodology can materially affect measured performance. RPMs inherit this problem in a more direct form because the benchmark may itself determine financial settlement.

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28. Data Revisions

Some measurements are not final when first published.

Economic statistics are a clear example.

A preliminary value may later be revised.

Suppose:

$$S_i(t) = 104$$

based on a preliminary release.

Months later the source revises the underlying data and the historical calculation becomes:

$$S_i(t) = 102.$$

Which value should settle the market?

Possible rules include:

- first published value;
- latest available value at settlement;
- value after a fixed revision window;
- designated final release;
- methodology-specific vintage.

The correct choice depends on the market's purpose.

But it must be specified **before participants trade**.

Otherwise the operator retains discretionary power over settlement.

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29. Goodhart's Law and Strategic Subjects

A particularly important problem arises when the subjects being measured can react to the measurement system.

Once a performance metric becomes economically consequential, subjects may optimize for the metric rather than the underlying phenomenon it was intended to represent.

Relative-performance research already warns that comparative evaluation can alter strategic incentives. Gibbons and Murphy discuss sabotage and collusion incentives arising under some relative-performance arrangements. (nber.org) Other work similarly shows that competitive relative-performance structures can affect disclosure and strategic behavior. (nber.org)

An RPM can therefore change the world it measures.

If billions of dollars eventually depend on a benchmark, participants or measured entities may attempt to:

- manipulate inputs;
- alter reporting methodology;
- delay unfavorable information;
- accelerate favorable information;
- influence benchmark membership;
- attack competing entities;
- influence the measurement institution itself.

This means an RPM is not merely a passive measurement system at scale.

It can become an incentive system.

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30. Liquidity Is Not Solvency

Another distinction is necessary.

A fully collateralized market may be solvent while still being illiquid.

Suppose every claim is backed by sufficient collateral.

A holder nevertheless wants to exit immediately.

If redemption requires another participant to purchase the position, the market can become illiquid despite remaining solvent.

Conversely, an automated redemption mechanism may provide immediate exits but become insolvent if its redemption formula does not preserve the rights of remaining participants.

Therefore:

$$\text{solvency} \neq \text{liquidity}.$$

A complete RPM must solve both.

Closed-pool conservation addresses the source of payouts.

It does not automatically provide efficient continuous exit.

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31. Mass Redemption

The most dangerous test for a closed RPM may not be ordinary trading.

It may be mass redemption after substantial redistribution.

Suppose one entity dramatically outperforms and its claims acquire a large share of the pool's redeemable value.

If most winning holders redeem simultaneously, the system must ensure that remaining claims retain exactly the collateral to which they are entitled.

Let:

$$C_t$$

be collateral immediately before redemption.

If participant u has a valid redeemable claim:

$$v_u,$$

then after redemption:

$$C_{t+1} = C_t - v_u.$$

But the remaining aggregate liabilities must also satisfy:

$$\sum_{j \neq u} V_j(t+1) = C_{t+1}.$$

This must remain true through arbitrary sequences of:

- deposits;
- withdrawals;
- reallocations;
- measurement updates;

- partial exits;
- epoch settlement.

A mechanism that cannot prove this invariant is not production-ready regardless of how intuitive its economic story appears.

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32. Manipulation and Economic Security

Traditional software security asks whether an attacker can violate program rules.

RPM security must additionally ask whether following the rules can still produce profitable manipulation.

Suppose manipulating an external measurement costs:

$$\$100,000.$$

Suppose that manipulation allows an attacker to earn:

$$\$5,000,000$$

from RPM positions.

The system is economically insecure even if every line of software executes correctly.

A rough security condition is therefore:

$$\text{Cost of manipulation} > \text{maximum realizable profit from manipulation.}$$

This becomes difficult in interconnected relative markets because manipulating one benchmark constituent may affect many claims simultaneously.

Economic security must therefore be considered at the **system level**, not merely per data source or per market.

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33. Relative Performance as a New Underlying

The deeper proposition of this paper can now be stated.

Financial markets traditionally trade:

- ownership;

- debt;
- commodities;
- currencies;
- rates;
- volatility;
- credit;
- event outcomes;
- externally measured variables.

Relative Performance Markets propose another category:

comparative measured performance.

The claim does not need to represent ownership of the measured entity.

It need not represent an absolute prediction.

It represents exposure to:

$$\text{entity performance} - \text{reference performance}.$$

This is economically meaningful because many real decisions are already comparative.

Capital allocation is comparative.

Competitive performance is comparative.

Benchmark evaluation is comparative.

Opportunity cost is comparative.

The proposed innovation is to make that comparison **native to the financial contract** rather than leaving participants to synthesize it through multiple independent positions.

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34. Is This Truly a New Primitive?

Caution is necessary when making novelty claims.

Nearly every component has precedent.

Prediction markets trade expectations.

Arrow-Debreu securities formalize state-contingent claims.

Derivatives create exposure to external variables.

Economic derivatives have directly traded macroeconomic releases. (nber.org)

Pari-mutuel markets mutualize payouts across participants. (sciencedirect.com)

Relative-performance evaluation compares subjects against peers or benchmarks. (nber.org)

Market scoring rules automate information markets. (ubplj.org)

Benchmark-relative trading already exists throughout portfolio management.

Therefore the novelty claim should not be:

> No one has ever traded relative performance.

That would be too broad.

The more defensible research proposition is:

> **A Relative Performance Market combines continuous comparative measurement, benchmark-relative contingent claims, expectation-driven price discovery, and potentially conserved multi-entity collateral into a unified market architecture.**

Whether that combination constitutes a distinct financial primitive ultimately depends on whether the resulting mechanism has properties that cannot be reduced economically to a trivial portfolio of existing instruments.

That question requires further formal work.

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35. Research Hypotheses

The framework produces several testable hypotheses.

H1 — Information aggregation

RPM prices can aggregate dispersed information about future comparative performance.

H2 — Common-shock filtering

Benchmark-relative measurement reduces sensitivity to shocks common to the measured population.

H3 — Natural opposition

Multi-entity relative measurement reduces the need for explicitly manufactured short counterparties because relative outperformance necessarily implies relative underperformance elsewhere.

H4 — Crowding sensitivity

Under conserved collateral, returns depend jointly on measured performance and existing capital distribution.

H5 — Benchmark sensitivity

Market outcomes are materially dependent on benchmark methodology.

H6 — Network information

Information about one entity can rationally alter the prices of other entities through the common benchmark.

H7 — Manipulation propagation

A common benchmark can amplify the financial consequences of corrupted measurements by propagating them across otherwise unrelated claims.

H8 — Comparative efficiency

For some classes of views, an RPM may express relative expectations with less capital or fewer independent positions than constructing equivalent exposure through separate absolute markets.

These propositions should be tested rather than assumed.

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36. Open Mathematical Problems

This paper intentionally does not present a final RPM mechanism.

Several problems require formal treatment.

36.1 Invariant-preserving redistribution

Find a function:

$$\mathcal{R}(A, S)$$

mapping allocations and relative performance into claim values such that:

$$\sum_i V_i = C$$

under all valid states.

36.2 Dynamic entry

New collateral must enter without acquiring previously accumulated performance.

36.3 Dynamic exit

Redemption must not transfer value improperly from remaining participants.

36.4 Switching

Changing exposure from A to B must be equivalent to a fair exit from A followed by fair entry into B, subject to fees and market impact.

36.5 Path dependence

Determine whether two identical terminal performance states reached through different paths should produce identical claim values.

36.6 Benchmark construction

Determine which benchmark properties produce economically meaningful relative measurement.

36.7 Price formation

Separate deterministic reference revaluation from expectation-driven secondary-market price discovery.

36.8 Liquidity

Provide continuous or sufficiently reliable liquidity without introducing hidden unbounded underwriting.

36.9 Settlement

Define finality where underlying measurements can be revised.

36.10 Manipulation bounds

Relate maximum economically extractable value to the cost of corrupting measurement inputs.

These are not secondary implementation details.

They determine whether Relative Performance Markets are economically coherent.

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37. Proposed Experimental Program

Before deploying real capital, RPMs should be studied experimentally.

A minimal simulation could contain:

$$N = 4$$

entities and:

$$C = \$100.$$

The simulation should test:

1. equal initial allocation;
2. concentrated initial allocation;
3. one entity outperforming;
4. all entities experiencing a common shock;
5. new participants entering after redistribution;
6. participants switching allocations;
7. winning participants partially redeeming;
8. mass redemption;
9. benchmark membership changes;
10. corrupted measurements;
11. delayed measurements;
12. revised measurements;
13. extreme concentration in one entity;
14. repeated oscillation in relative performance.

The objective should not initially be profitability.

The objective should be discovering whether the mechanism preserves its invariants and produces economically interpretable outcomes.

Only after these properties survive adversarial simulation should implementation complexity be introduced.

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38. Regulatory Characterization

Relative Performance Markets may be economically novel without being legally outside existing categories.

Depending on jurisdiction, implementation and underlying reference, an RPM could potentially be characterized as a derivative, swap, event contract, wagering product, security, or another regulated financial instrument.

Calling the mechanism a new "primitive" does not determine its legal classification.

Likewise, decentralization or use of cryptographic infrastructure does not inherently remove financial regulation.

Any real implementation would require jurisdiction-specific legal analysis separate from the economic analysis presented here.

This paper therefore uses **Relative Performance Market** as an economic and mechanism-design term rather than a proposed regulatory category.

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39. Limitations

Several limitations should be explicit.

First, relative performance is only as meaningful as the benchmark against which it is measured.

Second, normalized comparison can conceal important absolute deterioration. If every entity performs badly, some must still relatively outperform.

Third, measurement methodologies introduce assumptions that may be disputed.

Fourth, closed-pool systems may conserve collateral at the cost of making returns highly dependent on crowd positioning.

Fifth, a benchmark connecting many entities creates correlated oracle and manipulation risk.

Sixth, prediction-market literature does not guarantee that participants will efficiently aggregate complicated relative information.

Seventh, a continuously tradable market requires a liquidity mechanism that this paper has not solved.

Eighth, a conserved redistribution function that handles arbitrary deposits, withdrawals, switches, and settlement has not yet been formally derived here.

Ninth, a relative-performance claim may in some implementations be economically replicable using existing instruments, reducing the practical novelty of a dedicated market.

Finally, the usefulness of the primitive ultimately depends on whether there are domains where the relative measurement is both credible and valuable enough to attract sustained trading.

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40. Discussion

The significance of Relative Performance Markets is not that markets have never compared two things.

They have.

Nor is it that financial contracts have never settled against data.

They have.

Nor that mutualized pools, prediction markets, benchmark-relative investments, or contingent claims are individually novel.

They are not.

The interesting possibility lies in combining these ideas around a different organizing principle.

Instead of constructing a collection of isolated propositions:

> A rises.
>
> B rises.
>
> C falls.

an RPM constructs a common comparative environment:

> A relative to the system.
>
> B relative to the system.
>
> C relative to the system.

This changes the topology of the market.

The entities cease to be independent contracts and become members of a **relative-performance network**.

An update to one entity may affect the benchmark.

A benchmark change affects the relative state of other entities.

Capital allocation affects the economics of subsequent redistribution.

Market prices encode expectations about future relative states.

Settlement reconciles those expectations against measured outcomes.

The resulting system lies somewhere between an information market, a derivatives market, a pari-mutuel market, and a benchmark-relative allocation system.

That ambiguity is not necessarily a weakness.

It may be evidence that the design occupies an underexplored intersection between existing mechanisms.

41. Conclusion

Prediction markets demonstrate that uncertain future outcomes can be transformed into tradable contingent claims and that market prices can, under appropriate conditions, aggregate dispersed information. Derivatives demonstrate that exposure can be created around externally measured variables without requiring ownership of a conventional underlying asset. Economic derivatives have shown specifically that macroeconomic data itself can serve as the reference for financial contracts. Pari-mutuel systems demonstrate that risk can be mutualized across a pool rather than placed entirely on a dedicated counterparty. Relative-performance research demonstrates the economic significance of evaluating outcomes against peers and common benchmarks.

Relative Performance Markets combine these ideas around a simple proposition:

> **Performance does not have to be traded in isolation. Performance relative to a defined world can itself be the underlying.**

Formally, the primitive begins with:

$$R_i(t) = G(S_i(t), B(t)).$$

The challenge is then to construct financial claims around R_i that permit price discovery, continuous participation, fair entry and exit, objective settlement, sufficient liquidity, and—where closed collateral is desired—strict conservation:

$$\sum_i V_i(t) = C_t.$$

The conceptual case is straightforward.

The mathematical case is not yet complete.

That distinction is important.

The purpose of this paper is therefore not to claim that a production-ready Relative Performance Market has been solved. It establishes the proposed primitive, identifies its intellectual ancestry, derives its central properties, and exposes the mechanism-design problems that must be solved for the idea to become financially credible.

The next research question is no longer:

> Can relative performance be measured?

It is:

Can relative performance be transformed into a continuously tradable, liquid, manipulation-resistant and fully conserved market without requiring an external party to underwrite its aggregate outcomes?

The answer to that question determines whether Relative Performance Markets remain an interesting conceptual extension of prediction markets and derivatives—or become a distinct class of financial market.

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